Math 208 - Tutorial 4

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Distributing a Product, Question 3.4

The Lincoln Lock Company manufactures a commercial security lock at plants in Atlanta, Louisville, Detroit, and Phoenix. The unit cost of production at each plant is \$35.50, \$37.50, \$37.25, and \$36.25, and the annual capacities are 18,000, 15,000, 25,000, and 20,000, respectively. The locks are sold through wholesale distributors in seven locations around the country. The unit shipping cost for each plant–distributor combination is shown in the following table, along with the forecasted demand from each distributor for the coming year.

Unit Shipping Costs and Demand(Question 3.4)

(To) Distributor	**Tacoma**	**San Diego**	**Dallas**	**Denver**	**St. Louis**	**Tampa**	**Baltimore**
(From) Plant							
Atlanta	\$2.50	\$2.75	\$1.75	\$2.00	\$2.10	\$1.80	\$1.65
Louisville	\$1.85	\$1.90	\$1.50	\$1.60	\$1.00	\$1.90	\$1.85
Detroit	\$2.30	\$2.25	\$1.85	\$1.25	\$1.50	\$2.25	\$2.00
Phoenix	\$1.90	\$0.90	\$1.60	\$1.75	\$2.00	\$2.50	\$2.65
Demand	5,500	11,500	10,500	9,600	15,400	12,500	6,600

Table 1: Unit Shipping Costs (\$) and Forecasted Demand (units)

Questions

(a) Determine the least costly way of shipping locks from plants to distributors.

(b) Show the network diagram corresponding to the solution in (a). That is, label each of the arcs in the solution and verify that the flows are consistent with the given information.

(c) Suppose that the unit cost at each plant were \$10 higher than the original figure. What change in the optimal distribution plan would result? What general conclusions can you draw for transportation models with nonidentical plant-related costs?

Solution

Problem Decision Variables

The i corresponds to the plants, where $i = \{1, 2, 3, 4\}$, representing Atlanta, Louisville, Detroit, and Phoenix. Similarly, j corresponds to the distributors, where $j = \{1, 2, \ldots, 7\}$, representing Tacoma, San Diego, Dallas, Denver, St. Louis, Tampa, and Baltimore.

Problem Constraints

Supply Constraints (Plant Capacity):

$$\sum_{j=1}^{7} x_{ij} \leq C_i \quad \text{(capacity of plant } i\text{)}, \quad \forall i \in \{1, 2, 3, 4\}.$$

Demand Constraints (Distributor Requirement):

$$\sum_{i=1}^{4} x_{ij} \geq D_j \quad \text{(demand of distributor } j\text{)}, \quad \forall j \in \{1, 2, \ldots, 7\}.$$

Non-Negativity:

$$x_{ij} \geq 0, \quad \forall i, j.$$

Problem Objective

The objective function is to minimize the total cost, which includes the shipping costs associated with x_{ij}, the quantity shipped from plant i to distributor j. p_i is the production cost per unit at plant i.

$$\text{Minimize } Z = \sum_{i=1}^{4} \sum_{j=1}^{7} (p_i + c_{ij}) x_{ij}$$

Full Model

$$\min \sum_{i=1}^{4} \sum_{j=1}^{7} (p_i + c_{ij}) x_{ij}$$

Subject To:

$$\sum_{j=1}^{7} x_{ij} \leq C_i, \quad \forall i \in \{1, 2, 3, 4\},$$

$$\sum_{i=1}^{4} x_{ij} \geq D_j, \quad \forall j \in \{1, 2, \ldots, 7\}.$$

$$x_{ij} \geq 0, \quad \forall i, j.$$

The Goodwin Manufacturing Company Shipping Problem, 3.2(adapted)

The Goodwin Manufacturing Company is planning next week's shipments from its three manufacturing plants to its four distribution warehouses and is seeking a minimum-cost shipping schedule. Each plant has a potential capacity, expressed in cartons of product, and each warehouse has a demand requirement for the week that must be met. There are 12 possible shipment routes, and for every plant–warehouse combination, the unit shipping cost is known. The following table provides the given information.

Given Information

(To) Warehouse	**Atlanta**	**Boston**	**Chicago**	**Denver**	**Capacity**
(From) Plant					
Minneapolis	\$0.60	\$0.56	\$0.22	\$0.40	10,000
Pittsburgh	\$0.36	\$0.30	\$0.28	\$0.58	15,000
Tucson	\$0.65	\$0.68	\$0.55	\$0.42	15,000
Requirement	8,000	10,000	12,000	9,000	

Table 2: Shipping costs, capacities, and requirements

With the added elements:

- A factory can produce above capacity (overproduction), but at most $1.5\times$ its capacity.
- If a factory produces over capacity, it is penalized by a fixed cost of 5000.

Solution

Problem Parameters

Capacity and covering values, routes costs, overproduction allowance, and penalty. Let $r_{i,j}$, $i \in$ source, $j \in$ sink, be the cost of the route from source i to sink j. Let c_i, $i \in$ source, be the capacity of source i and d_j, $j \in$ sink, be the demand of sink j. Let o be the overproduction multiplier and let p be the penalty for overproduction.

Problem Decision Variables

One value per route, encoding the quantity flowing along the route. Let $f_{i,j}$ be the flow sent along the route from source i to sink j.

Problem Constraints

- **Capacity:** The sum of the flows leaving a source must be less than or equal to its capacity:

$$\sum_{j \in \text{sinks}} f_{i,j} \le oc_i, \quad \forall i \in \text{sources}.$$

- **Covering:** The sum of the flows entering a sink must be greater than or equal to its covering:

$$\sum_{i \in \text{sources}} f_{i,j} \ge d_j, \quad \forall j \in \text{sinks}.$$

- **Route capacities:** A natural variation on this model would also have route capacities (parameters) that would need to be satisfied (one constraint per route: flow $\le$ capacity).

Problem Objective

Minimize the sum over all routes of the product of the route cost (parameter) and route quantity (decision) and the penalty for overproduction:

$$\min \sum_{i \in \text{sources}} \sum_{j \in \text{sinks}} r_{i,j} f_{i,j} - p \sum_{i \in \text{sources}} \begin{cases} 1, & \text{if } \sum_{j \in \text{sinks}} f_{i,j} > c_i, \\ 0, & \text{otherwise.} \end{cases}$$

Full Model

$$\min \sum_{i \in \text{sources}} \sum_{j \in \text{sinks}} r_{i,j} f_{i,j} - p \sum_{i \in \text{sources}} \begin{cases} 1, & \text{if } \sum_{j \in \text{sinks}} f_{i,j} > c_i, \\ 0, & \text{otherwise.} \end{cases}$$

Subject To:

$$\sum_{j \in \text{sinks}} f_{i,j} \le oc_i, \quad \forall i \in \text{sources},$$

$$\sum_{i \in \text{sources}} f_{i,j} \ge d_j, \quad \forall j \in \text{sinks}.$$

Results

The shipping quantities are as follows: From Minneapolis, 2,640 cartons are shipped to Chicago, and 1,200 cartons are shipped to Denver. From Pittsburgh, 2,880 cartons are shipped to Atlanta, and 3,000 cartons are shipped to Boston. From Tucson, 2,520 cartons are shipped to Denver. No shipments are made to other destinations or from other plants beyond those specified.

The optimal minimized cost is \$22240.